UNIT I

Introduction to OR & Linear Programming

Handwritten study notes | B.Tech | Operations Research

Syllabus topics covered in this unit:

- Definition of OR, classification of models, modelling, methods of solving
- Limitations and applications of OR models
- LP: formulation, graphical method, simplex, Big-M, two-phase method
- Special cases: degeneracy, infeasibility, multiple optima, unboundedness
- Concept of duality and the dual theorems

1. Introduction to Operations Research

Origin: OR began in the Second World War (1939-40). British scientists (Blackett's 'Circus') were asked to find the best way of using limited radar, convoys and anti-aircraft resources. After the war the methods moved into industry, business and government. In India, OR started in 1949 at the Regional Research Laboratory, Hyderabad; the Operational Research Society of India was formed in 1957.

Definitions (learn any two)

Churchman, Ackoff & Arnoff (1957): OR is the application of scientific methods, techniques and tools to problems involving the operations of a system, so as to provide those in control of the system with optimum solutions to the problems.

Morse & Kimball: OR is a scientific method of providing executive departments with a quantitative basis for decisions regarding the operations under their control.

Characteristics / features of OR

- **Interdisciplinary team approach** - mathematicians, engineers, economists, statisticians work together
- **Scientific method** - observation, hypothesis, experiment, conclusion
- **Systems (holistic) approach** - the whole organisation is studied, not one department
- **Use of models** - real problem is represented by a mathematical model
- **Optimisation** - aim is the best (optimum) decision under given limits
- **Quantitative and computer based** - uses data, algorithms and software

Phases (steps) of an OR study

1. Define problem → 2. Build model → 3. Solve model → 4. Test validate → 5. Implement & control

- *Define the problem: objective, alternatives, limitations of the system*
- *Construct the model (mathematical equations or inequalities) from the data*
- *Solve the model with a suitable technique (simplex, MODI, etc.) to get a solution*
- *Validate - check whether results predict the real system correctly; update model*
- *Implement the solution and control it, since conditions keep changing*

2. Classification of OR Models

*A **model** is an idealised, simplified representation of a real-life situation. It shows the relation between the controllable (decision) variables, uncontrollable variables and the objective.*

Basis of classification	Types	Example
By structure	Iconic / Analogue / Symbolic	Toy plane; graph; equation
By nature of environment	Deterministic / Probabilistic	LP model; queueing model
By time	Static / Dynamic	EOQ; dynamic programming
By purpose	Descriptive / Predictive / Prescriptive	Survey; forecast; LP
By method of solution	Analytical / Simulation	Calculus; Monte-Carlo
By generality	General / Specific	Simplex; sales-forecast

- *Iconic model: physical look-alike, scaled up or down (globe, scale model of a dam, aircraft)*
- *Analogue model: one property stands for another (contour map, thermometer, graphs, speedometer)*
- *Symbolic (mathematical) model: letters and symbols for variables, equations for relations.* ***Most common in OR.***
- *Deterministic: all data known with certainty. Probabilistic (stochastic): data follow probability laws*
- *Static: decision for one period only. Dynamic: decisions are linked over several periods*

Modelling - qualities of a good model

- *Simple enough to be solved, yet realistic enough to be useful*
- *Easy to modify and update; data should be easily available*
- *Includes only the **important** variables, ignoring minor details*
- *Steps: identify variables -> state objective -> write constraints -> solve -> test & revise*

Methods of solving OR models

- *Analytical method: classical mathematics (calculus, algebra, matrices) gives a direct formula, e.g. EOQ*
- *Iterative (numerical) method: start with a trial solution, improve step by step till the optimum, e.g. simplex, MODI*
- *Monte-Carlo (simulation) method: random sampling is used to imitate the system when equations are too complex*
- *Heuristic method: rules of thumb that give a good (not always optimal) answer quickly*

3. Limitations and Applications of OR

Limitations

- Models are only approximations; real life has factors that cannot be quantified (human feelings)
- Collecting accurate data is costly and time-consuming; wrong data gives wrong decisions
- Large problems need expensive computers and trained experts
- OR gives a **recommendation**, it cannot replace managerial judgement
- Managers may resist change; implementation may fail even if the model is correct

Applications

Area	Typical use of OR
Production / Inventory	Product mix, scheduling, EOQ, plant location
Marketing	Media selection, pricing, sales-force allocation
Finance	Capital budgeting, portfolio selection, cash flow
Transportation	Least cost distribution, routing, assignment of crews
Personnel	Manpower planning, recruitment, shift scheduling
Defence / Govt / Health	Resource allocation, hospital queues, project planning

Main OR techniques: LP, transportation, assignment, game theory, queuing, inventory, replacement, sequencing, dynamic programming, simulation, PERT/CPM.

4. Linear Programming (LP) - Formulation

LP is a mathematical technique for allocating **limited resources** among competing activities so that a linear objective function is optimised (maximised profit or minimised cost) subject to linear constraints.

Assumptions of LP

- *Linearity / proportionality* - contribution and resource use are proportional to the quantity
- *Additivity* - total = sum of individual parts *Divisibility* - fractional values allowed
- *Certainty* - all coefficients are known constants *Non-negativity* - variables are $x_j \geq 0$

General form of an LP model

Maximise (or Minimise) $Z = c_1x_1 + c_2x_2 + \ldots + c_nx_n$ *(objective function)*

subject to $a_{i1}x_1 + a_{i2}x_2 + \ldots + a_{in}x_n \ (\leq , = , \geq) \ b_i, \quad i = 1, 2, \ldots, m$ *(constraints)*

$x_1, x_2, \ldots, x_n \geq 0$ *(non-negativity)*

Terms to remember

- **Feasible solution:** satisfies all constraints. **Optimal solution:** feasible and gives best Z
- **Basic solution:** n - m variables set to zero, rest solved. **BFS:** basic solution that is also feasible
- **Feasible region:** common area of all constraints; it is a **convex set**

Steps in formulation

(1) Identify decision variables (2) write objective function (3) list resource constraints from the data (4) add non-negativity.

Example 1 - Product mix (maximisation)

A firm makes products P_1 and P_2. P_1 needs 2 hrs on machine A and 4 hrs on machine B per unit; P_2 needs 3 hrs on A and 3 hrs on B. Machine A has 60 hrs and B has 96 hrs available. Profit is Rs 40 per unit of P_1 and Rs 35 per unit of P_2. Formulate the LP.

Let x_1, x_2 = number of units of P_1, P_2 produced.
Maximise $Z = 40x_1 + 35x_2$
subject to $2x_1 + 3x_2 \le 60$ (machine A)
$4x_1 + 3x_2 \le 96$ (machine B)
$x_1, x_2 \ge 0$

Example 2 - Diet problem (minimisation)

Food A costs Rs 6/unit and gives 2 units protein, 1 unit vitamin. Food B costs Rs 4/unit and gives 1 unit protein, 3 units vitamin. Daily need: at least 12 protein and 15 vitamin. Minimise cost.

x_1, x_2 = units of food A, B. Minimise $Z = 6x_1 + 4x_2$
s.t. $2x_1 + x_2 \ge 12$ (protein), $x_1 + 3x_2 \ge 15$ (vitamin), $x_1, x_2 \ge 0$

Example 3 - Blending / investment type

Rs 1,00,000 is to be invested in bonds (x_1, return 8%) and shares (x_2, return 12%). Not more than 60% in shares and at least 20% in bonds.

Max $Z = 0.08x_1 + 0.12x_2$ s.t. $x_1 + x_2 \le 1{,}00{,}000$; $x_2 \le 60{,}000$; $x_1 \ge 20{,}000$; $x_1, x_2 \ge 0$

Standard form - slack, surplus and artificial variables

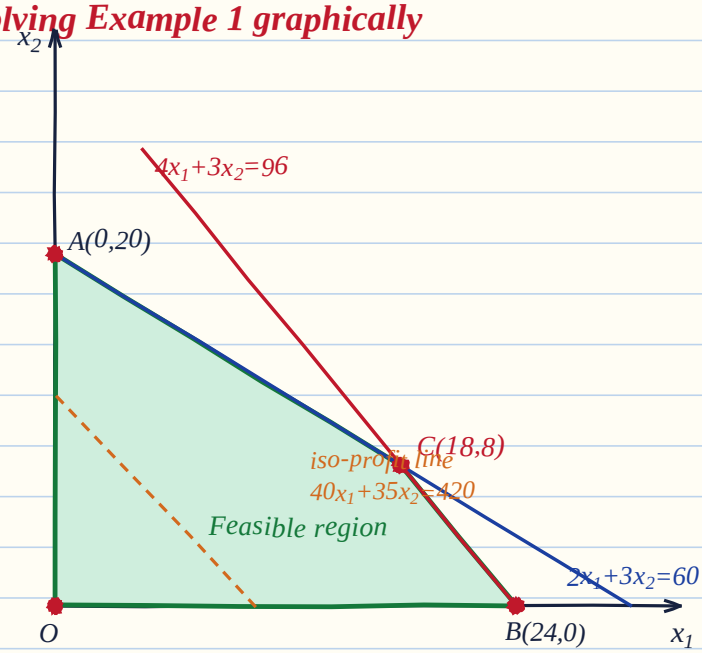
Constraint type	Add / subtract	Cost in Z (Max)	Cost in Z (Min)	Starting basic?
≤	+ slack s	0	0	Yes (s)
≥	- surplus s, + artificial a	s: 0, a: -M	s: 0, a: +M	Yes (a)
=	+ artificial a	-M	+M	Yes (a)

Rule: RHS of every constraint must be non-negative. If $b < 0$ multiply the whole constraint by -1 (inequality reverses). Slack = unused resource; surplus = excess over the minimum.

5. Graphical Method (two variables)

- Step 1: convert each inequality to an equation and plot the line
- Step 2: shade the side satisfying the inequality; the common area is the **feasible region**
- Step 3: find the coordinates of all corner (extreme) points
- Step 4: compute Z at each corner point - best value is optimal (**corner point theorem**), or move an iso-profit/iso-cost line parallel to itself

Solving Example 1 graphically



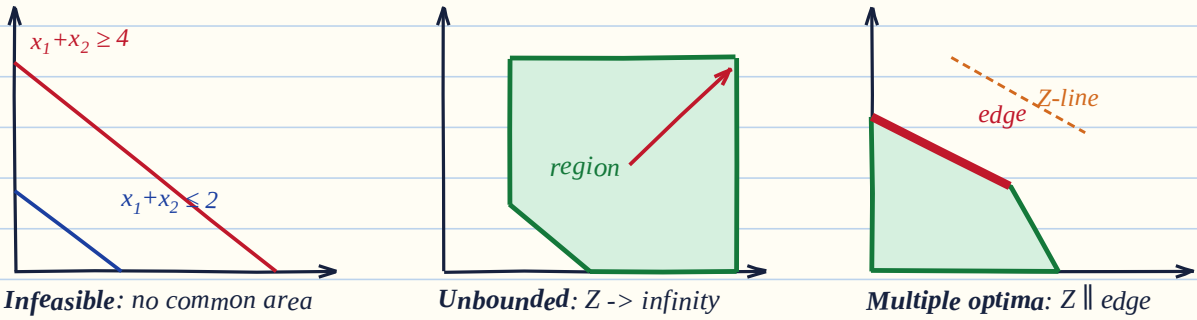
Corner pt	$Z = 40x_1+35x_2$
O (0,0)	0
B (24,0)	960
C (18,8)	**1000 <- max**
A (0,20)	700

Optimum: $x_1=18$, $x_2=8$

Max Z = Rs 1000

C = intersection of both lines:
subtract: $2x_1=36 \Rightarrow x_1=18$, $x_2=8$

Special cases seen on the graph



Infeasible: no common area — **Unbounded**: Z -> infinity — **Multiple optima**: Z ∥ edge

Note: multiple optimum = whole edge optimal, so infinitely many optimal solutions. Check by hand: Max $Z = 2x_1+4x_2$; $x_1+2x_2 \leq 5$; $x_1+x_2 \leq 4$ gives $Z = 10$ at both (0, 2.5) and (3, 1).

6. Simplex Method

Used when there are more than two variables. It moves from one basic feasible solution (corner) to an adjacent better one till no improvement is possible.

Algorithm (maximisation)

- Step 1: write standard form (add slacks); make all $b_i \geq 0$. Initial BFS: slacks in the basis
- Step 2: build the table; compute $Z_j = \sum C_B \cdot a_j$ and $C_j - Z_j$ (net evaluation row)
- Step 3: **Optimality test** - if all $(C_j - Z_j) \leq 0$ (Max) the solution is optimal. For Min: all ≥ 0
- Step 4: **Entering variable** = largest positive $(C_j - Z_j)$ -> key column

- Step 5: **Leaving variable** = minimum ratio b_i / (key-column entry > 0) -> key row; pivot = intersection
- Step 6: new pivot row = old row / pivot; other rows: $R_{new} = R_{old}$ - (key column entry)×(new pivot row). Repeat from Step 2

Solving Example 1 by simplex

Standard form: Max $Z = 40x_1 + 35x_2 + 0s_1 + 0s_2$; $2x_1+3x_2+s_1 = 60$; $4x_1+3x_2+s_2 = 96$.

Iteration 0

	Cj	40	35	0	0		
Cb	**Basis**	**x1**	**x2**	**s1**	**s2**	**b**	**Ratio**
0	s1	2	3	1	0	60	30
0	s2	4	3	0	1	96	24
	Zj	0	0	0	0	0	
	Cj-Zj	40	35	0	0		

Entering = **x1**, Leaving = **s2**, Pivot element = **4**

Iteration 1

	Cj	40	35	0	0		
Cb	**Basis**	**x1**	**x2**	**s1**	**s2**	**b**	**Ratio**
0	s1	0	3/2	1	-1/2	12	8
40	x1	1	3/4	0	1/4	24	32
	Zj	40	30	0	10	960	
	Cj-Zj	0	5	0	-10		

Entering = **x2**, Leaving = **s1**, Pivot element = **3/2**

Final table (iteration 2)

	Cj	40	35	0	0		
Cb	**Basis**	**x1**	**x2**	**s1**	**s2**	**b**	**Ratio**
35	x2	0	1	2/3	-1/3	8	
40	x1	1	0	-1/2	1/2	18	
	Zj	40	35	10/3	25/3	1000	
	Cj-Zj	0	0	-10/3	-25/3		

All Cj - Zj satisfy the optimality condition, so STOP. Optimal reached.

Result: $x_1 = 18, x_2 = 8, s_1 = s_2 = 0$, Max $Z = 1000$ (same as graphical).
Shadow prices (Z_j under slacks): machine A = 10/3, machine B = 25/3 per extra hour.

7. Big-M Method (Charne's penalty method)

Used when constraints are of type ≥ or = , because there is no obvious starting basis. Artificial variables are added and given a huge penalty M (M is larger than any number in the problem).

- Min problem: artificial cost = +M. Max problem: artificial cost = -M
- Solve like simplex. Once an artificial leaves the basis, its column may be dropped
- Optimal with an artificial still in basis at positive level => problem is INFEASIBLE

Example 4: Min $Z = 4x_1 + x_2$; $3x_1+x_2 = 3$; $4x_1+3x_2 \geq 6$; $x_1+2x_2 \leq 4$

Standard form: $3x_1+x_2+a_1 = 3$; $4x_1+3x_2-s_2+a_2 = 6$; $x_1+2x_2+s_3 = 4$. Objective: Min $Z = 4x_1+x_2+M\,a_1+M\,a_2$.

Table 1

	Cj	4	1	M	0	M	0		
Cb	**Basis**	**x1**	**x2**	**a1**	**s2**	**a2**	**s3**	**b**	**Ratio**
M	a1	3	1	1	0	0	0	3	1
M	a2	4	3	0	-1	1	0	6	3/2
0	s3	1	2	0	0	0	1	4	4
	Zj	7M	4M	M	-M	M	0	9M	
	Cj-Zj	4 - 7M	1 - 4M	0	M	0	0		

Entering = **x1**, Leaving = **a1**, Pivot element = **3**

Table 2

	Cj	4	1	M	0	M	0		
Cb	**Basis**	**x1**	**x2**	**a1**	**s2**	**a2**	**s3**	**b**	**Ratio**
4	x1	1	1/3	1/3	0	0	0	1	3
M	a2	0	5/3	-4/3	-1	1	0	2	6/5
0	s3	0	5/3	-1/3	0	0	1	3	9/5
	Zj	4	4/3 + 5/3M	4/3 - 4/3M	-M	M	0	4 + 2M	
	Cj-Zj	0	-1/3 - 5/3M	-4/3 + 7/3M	M	0	0		

Entering = **x2**, Leaving = **a2**, Pivot element = **5/3**

Table 3

	Cj	4	1	M	0	M	0		
Cb	**Basis**	**x1**	**x2**	**a1**	**s2**	**a2**	**s3**	**b**	**Ratio**
4	x1	1	0	3/5	1/5	-1/5	0	3/5	3
1	x2	0	1	-4/5	-3/5	3/5	0	6/5	
0	s3	0	0	1	1	-1	1	1	1
	Zj	4	1	8/5	1/5	-1/5	0	18/5	
	Cj-Zj	0	0	-8/5 + M	-1/5	1/5 + M	0		

*Entering = **s2**, Leaving = **s3**, Pivot element = **1***

Table 4

	Cj	4	1	M	0	M	0		
Cb	**Basis**	**x1**	**x2**	**a1**	**s2**	**a2**	**s3**	**b**	**Ratio**
4	x1	1	0	2/5	0	0	-1/5	2/5	
1	x2	0	1	-1/5	0	0	3/5	9/5	
0	s2	0	0	1	1	-1	1	1	
	Zj	4	1	7/5	0	0	-1/5	17/5	
	Cj-Zj	0	0	-7/5 + M	0	M	1/5		

All Cj - Zj satisfy the optimality condition, so STOP. Optimal reached.

Optimal: $x_1 = 2/5, x_2 = 9/5, s_2 = 1$, Min Z = 17/5 = 3.4

Check: 3(0.4) + 1.8 = 3 (ok); 4(0.4)+3(1.8) = 7 ≥ 6 (ok); 0.4 + 3.6 = 4 (ok).

8. Two-Phase Simplex Method

Removes the use of M (avoids big numbers and computer round-off). The same problem is solved in two phases.

Phase I: replace objective by Min W = sum of artificial variables (cost 1 for each a, 0 for all others). Solve by simplex.

- if Min W = 0 -> a feasible solution exists, go to Phase II.
- if Min W > 0 -> problem is infeasible, stop.

Phase II: delete artificial columns, restore the ORIGINAL objective coefficients, and continue simplex from the final Phase I table.

Example 4 again by two-phase method

Phase I: Min W = $a_1 + a_2$

Phase I - table 1

	Cj	0	0	1	0	1	0		
Cb	**Basis**	**x1**	**x2**	**a1**	**s2**	**a2**	**s3**	**b**	**Ratio**
1	a1	3	1	1	0	0	0	3	1
1	a2	4	3	0	-1	1	0	6	3/2
0	s3	1	2	0	0	0	1	4	4
	Zj	7	4	1	-1	1	0	9	
	Cj-Zj	-7	-4	0	1	0	0		

*Entering = **x1**, Leaving = **a1**, Pivot element = **3***

Phase I - table 2

	Cj	0	0	1	0	1	0		
Cb	**Basis**	**x1**	**x2**	**a1**	**s2**	**a2**	**s3**	**b**	**Ratio**
0	x1	1	1/3	1/3	0	0	0	1	3
1	a2	0	5/3	-4/3	-1	1	0	2	6/5
0	s3	0	5/3	-1/3	0	0	1	3	9/5
	Zj	0	5/3	-4/3	-1	1	0	2	
	Cj-Zj	0	-5/3	7/3	1	0	0		

Entering = **x2**, Leaving = **a2**, Pivot element = **5/3**

Phase I - table 3

	Cj	0	0	1	0	1	0		
Cb	**Basis**	**x1**	**x2**	**a1**	**s2**	**a2**	**s3**	**b**	**Ratio**
0	x1	1	0	3/5	1/5	-1/5	0	3/5	
0	x2	0	1	-4/5	-3/5	3/5	0	6/5	
0	s3	0	0	1	1	-1	1	1	
	Zj	0	0	0	0	0	0	0	
	Cj-Zj	0	0	1	0	1	0		

All Cj - Zj satisfy the optimality condition, so STOP. Optimal reached.

Min W = 0 and no artificial is in the basis => feasible solution found. Go to Phase II.

Phase II: Min $Z = 4x_1 + x_2$ (artificial columns deleted)

Phase II - table 1

	Cj	4	1	0	0		
Cb	**Basis**	**x1**	**x2**	**s2**	**s3**	**b**	**Ratio**
4	x1	1	0	1/5	0	3/5	3
1	x2	0	1	-3/5	0	6/5	
0	s3	0	0	1	1	1	1
	Zj	4	1	1/5	0	18/5	
	Cj-Zj	0	0	-1/5	0		

Entering = **s2**, Leaving = **s3**, Pivot element = **1**

Phase II - table 2

	Cj	4	1	0	0		
Cb	**Basis**	**x1**	**x2**	**s2**	**s3**	**b**	**Ratio**
4	x1	1	0	0	-1/5	2/5	
1	x2	0	1	0	3/5	9/5	
0	s2	0	0	1	1	1	
	Zj	4	1	0	-1/5	17/5	
	Cj-Zj	0	0	0	1/5		

All Cj - Zj satisfy the optimality condition, so STOP. Optimal reached.

Same answer: $x_1 = 2/5, x_2 = 9/5, Z = 17/5$.

Point	Big-M	Two-Phase
Penalty	M inside the table (symbolic)	No M; two separate LPs
Work	Single table sequence	Phase I + Phase II
Risk	Round-off with large M	More tables, but safer
Infeasible test	Artificial positive in basis	Min W > 0 in Phase I

9. Special Cases in LP (simplex view)

Case	How to recognise in the table	Meaning
Degeneracy	Tie in minimum ratio; a basic variable = 0	Possible cycling
Infeasibility	Optimal table has artificial in basis with value > 0	No feasible region
Multiple optima	Non-basic variable has Cj-Zj = 0 at optimum	Infinitely many optima
Unboundedness	Entering column has no positive entry (no ratio)	Z -> infinity

(a) Degeneracy

A BFS is degenerate if one or more basic variables are zero. It arises when two rows tie for the minimum ratio; the next table then has a zero in b-column. Danger: cycling (same bases repeat). Cure: break the tie by **Bland's rule** (choose the lowest index) or by perturbing b-values slightly.

Example: Max $Z = 3x_1 + 9x_2$; $x_1+4x_2 \le 8$; $x_1+2x_2 \le 4$.
First ratios: 8/4 = 2 and 4/2 = 2 (tie!). x_2 enters; whichever row leaves, the other slack stays basic at ZERO => degenerate.
Optimum Z = 18 at $x_1 = 0, x_2 = 2$. In graph: three lines pass through one point (redundant constraint).

(b) Infeasibility

Example: Max $Z = 3x_1+2x_2$; $x_1+x_2 \le 2$; $x_1+x_2 \ge 4$. No point satisfies both. In Big-M, the final table keeps an artificial variable at a positive value ($a_2 = 2 > 0$).

(c) Multiple optimal solutions

When a non-basic variable has $C_j - Z_j = 0$ in the optimal table, entering it gives another corner with the same Z. If X^1, X^2 are optimal then $X = \lambda X^1 + (1-\lambda)X^2$, $0 \le \lambda \le 1$ is also optimal. (see example in section 5.)

(d) Unbounded solution

Example: Max $Z = 3x_1+2x_2$; $x_1-x_2 \le 1$; $3x_1+x_2 \ge 3$. The feasible region extends to infinity along x_2 and Z grows without limit. Wrong formulation in practice - re-check constraints.

10. Concept of Duality

Every LP (called **primal**) has a companion LP called its **dual**. Both use the same data (A, b, c). Solving one gives the solution of the other; the dual of the dual is the primal.

Primal (Max): $\text{Max } Z = c\,x \quad \text{s.t. } A\,x \le b,\ x \ge 0$

Dual (Min): $\text{Min } W = b^T y \quad \text{s.t. } A^T y \ge c^T,\ y \ge 0$

Rules for converting primal to dual

Primal (Max)	Dual (Min)
Objective coefficients c_j	RHS constants of dual constraints
RHS constants b_i	Objective coefficients of dual
i-th constraint $\le$	i-th dual variable $y_i \ge 0$
i-th constraint $\ge$	$y_i \le 0$ (or multiply by -1 first)
i-th constraint $=$	y_i unrestricted in sign
Variable $x_j \ge 0$	j-th dual constraint is $\ge$
Variable x_j unrestricted	j-th dual constraint is $=$
m constraints, n variables	n constraints, m variables

Example: dual of Example 2 (diet problem)

Primal: Min $Z = 6x_1+4x_2$; $2x_1+x_2 \ge 12$; $x_1+3x_2 \ge 15$; $x \ge 0$

Dual: Max $W = 12y_1 + 15y_2$; $2y_1 + y_2 \le 6$; $y_1 + 3y_2 \le 4$; $y_1, y_2 \ge 0$

Solving: primal $x_1 = 4.2$, $x_2 = 3.6$, $Z = 39.6$; dual $y_1 = 2.8$, $y_2 = 0.4$, $W = 39.6$ -> **$Z = W$**

Check with Example 1: its dual is Min $W = 60y_1+96y_2$; $2y_1+4y_2 \ge 40$; $3y_1+3y_2 \ge 35$. With $y = (10/3, 25/3)$: $W = 200+800 = 1000 = Z$. These y-values are the shadow prices read from the Z_j row.

Duality theorems

- *Weak duality:* for any feasible x (primal Max) and y (dual Min), $c\,x \le b^T y$
- *Strong (fundamental) duality:* if one problem has an optimal solution so does the other and Max Z = Min W
- *Existence theorem:* if primal is unbounded the dual is infeasible; if primal is infeasible the dual is unbounded or infeasible
- *Complementary slackness:* x_j·(slack of dual j-th constraint) = 0 and y_i·(slack of primal i-th constraint) = 0 at optimum

Why study the dual? / Economic meaning

- y_i = *shadow price*: change in optimal Z per unit increase in resource b_i
- If a resource has slack > 0, its shadow price is zero (not scarce)
- Dual may be easier: more constraints than variables -> solve dual instead (fewer rows)
- Basis of dual simplex method and sensitivity (post-optimality) analysis

11. Quick Revision & Important Questions

1. Define OR; list phases; classify models. 2. Limitations & applications of OR.
3. Formulate and solve by graphical method (corner points). 4. Simplex algorithm with example.
5. Big-M vs Two-phase method with problem. 6. Explain degeneracy, infeasibility, unbounded, multiple optima.
7. Write dual of a given LP; state duality theorems.

Formula sheet: $Z_j = \sum C_B\, a_{ij}$; Net evaluation = $C_j - Z_j$; Ratio = b_i/a_{ik} ($a_{ik} > 0$); Max optimal when all $C_j - Z_j \le 0$, Min optimal when all ≥ 0; Max Z = Min W.